

Aryaman Pattnayak

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Professional Summary — Machine Learning Engineer with expertise in building AI solutions, specializing in model fine-tuning, data synthesis, and complex AI pipeline orchestration. Skilled in developing explainable AI systems with a focus on transparency, traceability, and agentic workflows. Published author in bio-silicon intelligence and experienced technical writer with deep knowledge of AI research, cloud computing, and advanced machine learning.

Skills

Cloud Azure, AWS	Containerization Docker
Languages Python, Bash, C++ , Java, JavaScript, CUDA	Libraries NumPy, Pandas, OpenAI, Anthropic,
Frameworks PyTorch, TensorFlow, Transformers	DeepSpeed
OS Ubuntu, Windows	

Experience

StoicAI

January 2025 – Present

Machine Learning Engineer

- Developed **autonomous trading agents**, optimizing strategy execution and decision-making.
 - Created a **Twitter AI agent** for **automated tweeting and smart replies**, enhancing engagement.
 - **\$STOIC**(ModernStoic) reached an **all-time high market cap of \$25M USD** on Solana.
- [DexScreener Link](#)

Omega Labs

October 2024 – Present

Machine Learning Research Engineer

- **Primary engineer in charge of Subnet 21 and 24 of Bittensor**, overseeing the development, optimization, and maintenance of decentralized AI networks.
- Benchmarked **ImageBind+Llama** models on sn21.ai, creating benchmarking scripts and optimizations.
- Developed the **backend for sn21.ai**, including **Python relay servers** and infrastructure for efficient query handling.
- Processed data from **Omega Focus** screen recordings, generating **annotations** and summaries. Implemented **retrieval-augmented thinking** to provide hyper-specific insights into user activities.
- Created **Omega Forge**, a **reasoning API** supporting **Mixture of Agents (MoA)**, **Chain of Code (CoC)**, and **Monte Carlo Tree Search (MCTS)** for advanced AI reasoning.
- Built **Omega Twins**, hyper-personalized **social AI agents** leveraging screen recording annotations from Omega Focus.

Nous Research

Jul 2024 – Oct 2024

Machine Learning Research Engineer

- Contributed to Nous Forge, an AI orchestration platform designed to provide a transparent, auditable alternative to closed systems. Focused on building a modular, editable chain of thought for AI workflows, enabling users to trace, explain, and modify every decision step.
- Developed dynamic AI pipelines that enable seamless chaining of models and switching based on specific tasks for the user in Nous Forge.
- Led the AgentInstruct Project, focusing on automating data generation and validation using LLM-powered agents for the generation of high quality synthetic data to be used for the training of the future Hermes 4 models. Designed a data verification pipeline involving Judge, Reviewer, Suggester, and Editor Agents, ensuring data quality, diversity, and correctness. Created modular workflows for content transformation, seed instruction generation, and instruction refinement. Built an Agentic Component Library to support flexible integration, and developed workflows for training data across 17 skills, enhancing agent autonomy and adaptability. Additionally, designed and developed agents equipped with specialized tools such as interpreters, code executors, and web search capabilities entirely from scratch. These tools were independently built without relying on existing frameworks or libraries, providing a unique, custom solution that enhanced the agents' versatility and performance in diverse tasks.
- Contributed to Nous WorldSim, a multi-agent simulation environment blending real-world scenarios with imaginative simulations for AI training. Developed interactive simulations with Hermès LLM for virtual world-building, enhancing both research and entertainment. Improved user interaction on WorldSim, enabling collaborative world-building and community-driven experiences, and trained agents for negotiation, problem-solving, and decision-making in realistic environments.

Craftbench.ai

Jan 2024 – Jul 2024

Senior ML Engineer (LLM Engineer)

- Developed synthetic dataset creation software leveraging GPT-4 for flexible, scalable data generation. Enabled switching between LLM models, allowing for experimentation with different architectures.

- Created Craftsim, a simulate-any-game platform in ASCII art using LLM, designed to interact with game environments dynamically. This innovative project is scheduled for future release and is aimed at enhancing the user experience in interactive simulations.
- Built a highly efficient Discord scraper with a Rust backend, designed to generate QA pairs from threads. The scraper integrated OpenAI APIs and optimized data collection for model fine-tuning, resulting in a highly scalable data generation framework.
- Developed numerous finetuning scripts using PyTorch, Torch Compile, and TensorFlow, which significantly reduced the training time for large language models, particularly for creative content generation and question-answering tasks.
- Designed and implemented CraftDnD, the world's first AI-driven Dungeons and Dragons experience, using stable diffusion models and MJS code for interactive scene creation, showcasing advanced AI-augmented gameplay.
- Configured distributed GPU training using two H100 GPUs, enabling the fine-tuning of LLMs with over 70 billion parameters. Successfully integrated LoRA, QLoRA, and PEFT methods to fine-tune large models more efficiently.
- Fine-tuned 8 open-source LLMs on synthetic datasets for deployment in real-world applications, including a Minecraft question-answering bot used internally by the company.
- Implemented Retrieval-Augmented Generation (RAG) techniques, which reduced hallucinations in generated text by 95%, enhancing the factual accuracy of AI-generated content.
- Created Agentic Frameworks for automating code execution within terminal environments, revolutionizing user interactions with complex software systems, including automating Minecraft modding and texture generation.

AzO Network

Aug 2023 – Feb 2024

AI Technical Writer

- Authored over 100 AI-focused articles, targeting emerging AI trends and technological advancements. The content was designed to increase user engagement and provide in-depth insights into complex AI topics for a technical audience.
- Conducted extensive research on cutting-edge AI trends, ensuring high-quality, accurate, and timely content. Published articles ranged from technical overviews of AI models to detailed discussions on emerging AI applications.

Pyxeda.AI

Jan 2023 – Aug 2023

AI Consultant

- Trained 50+ students in advanced Python programming and machine learning concepts, focusing on practical applications and real-world problem-solving.
- Led multiple AI and ML bootcamps, educating over 200 participants on AI-driven solutions, including model development, data pre-processing, and model deployment.

Education

Shiv Nadar University

Bachelor of Technology in Computer Science and Engineering

Publications

4.1 Development of Bio-Silicon Synergetic Intelligence Systems

Phase 1: Bio-Silicon Synergetic Intelligence System

[ResearchGate Link](#)

- Conducted research on brain-computer interfaces (BCI) integrating human cortical organoids into rat brains, as part of developing advanced neuro-technological systems.
- Developed bio-silicon integration methods aimed at creating a bidirectional communication channel between brain matter and silicon-based computational systems. Tested using a DOOM gaming simulation for neural signal translation.
- Implemented microelectrode arrays (MEA) to facilitate neural signal transmission, allowing communication between rat brains and computational systems.
- Created AI-driven feedback loops for real-time neural signal interpretation, enhancing supervised learning processes for neuro-silicon interaction.

4.2 Advancements in Hybrid Computing Systems

Bio-Silicon Intelligence System

[ResearchGate Link](#)

- Focused on integrating analogue and digital computing in biological-silicon hybrid systems, using advanced carbon nanotube-coated electrodes for high-fidelity neural interfacing.
- Developed systems to process EEG signals and generate computational outputs based on continuous real-time data.
- Investigated biocomputing's potential in enhancing brain-like learning and memory processes in silicon-based systems.

4.3 AI Integration in Neuro-Silicon Platforms

AI-Driven Bio-Silicon Intelligence System

[ResearchGate Link](#)

- Developed an AI-driven platform integrating biological neural networks with silicon-based computing for advanced neuro-technological applications.

- Designed reinforcement learning models to optimize communication between biological brains and machine learning systems.
- Advanced the understanding of cortical organoid interactions with silicon computing systems, using real-time neural signal decoding to control machine operations in a simulated environment.

Extracurriculars & Achievements

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| ARC Challenge by Allen Institute for AI | Jan 2025 – Present |
| <ul style="list-style-type: none">– Achieved 1st position with GPT-4o+CoT approach, demonstrating advanced scientific reasoning capabilities for multiple-choice question answering. ARC Challenge Link | |
| OpenbookQA Challenge by Allen Institute for AI | Apr 2024 – Present |
| <ul style="list-style-type: none">– Achieved 1st position with 96.80% accuracy using advanced NLP techniques such as SentenceTransformers, KnowledgeBase, Chain-of-Thought prompting. OpenbookQA Link | |
| Pie and AI Ambassador at DeepLearning.AI | Nov 2023 – Present |
| <ul style="list-style-type: none">– Organized AI-focused events and led community growth initiatives, fostering the development of local AI communities and promoting AI literacy. | |
| Winner of SmartSNU Hackathon | Sep 2023 |
| <ul style="list-style-type: none">– Developed a phishing link detection system using a browser extension, leveraging machine learning techniques to classify and block malicious links in real-time. | |
| Founder and President of Analytica | Feb 2023 – Present |
| <ul style="list-style-type: none">– Founded the first AI/ML and Data Science Technical Society at Shiv Nadar University, organizing workshops, hackathons, and research talks to engage students in cutting-edge AI technologies and data science practices. | |